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This paper has developed a simple model of the spread of misinformation over social media platforms. A group of Bayesian agents with heterogeneous priors receive and share news items (articles) according to a stochastic sharing network, determined by the social media platform. Articles may be truthful and informative about an underlying state, or may contain misinformation, making them (weakly) anti-correlated with the underlying state. Upon receiving an article, an agent can decide to share it with others, ignore it, or actively call out another agent for propagating misinformation (“dislike”). Misinformation spreads when agents share articles expecting positive social media feedback and little negative reactions.

Though simple and parsimonious, the model encapsulates several rich strategic interactions. Agents receive utility from sharing truthful articles and not misinformation, but also enjoy peer engagement with shared content. The former force pushes agents toward sharing pro-attitudinal content, whereas the latter force pushes agents toward pleasing those in their sharing network. The ideological congruence between an agent and those in her sharing network, which we capture with the notion of homophily, is critical for sharing decisions. Because individuals are more likely to dissent against articles that disagree with their prior beliefs, an agent will be more cautious in sharing articles that disagree with the views of those in her sharing network.

We provide several comparative static results. Some of those are very intuitive, though still useful for interpreting a range of results in the emerging empirical literature on social media and misinformation. For example, we find that while misinformation typically spreads less than truthful content (holding all else constant), more sensational content tends to be shared more. Moreover, when misinformation is correlated with sensationalism, the rapid spread misinformation can be problematic.

Of particular interest are the comparative statics with respect to homophily—measuring whether agents tend to share articles with others with similar or dissimilar beliefs. We show that when there is a highly-reliable article, an increase in homophily reduces the virality of content. Because this article is unlikely to contain misinformation, it is of broad appeal to a wide range of social media users, independent of ideology. An increase in homophily then reduces the extent to which this article can spread throughout the sharing network. The implications of homophily for low-reliability articles are very different, however: in a well-connected network, such articles will be disliked and stopped by users who disagree with their message, and anticipating this behavior and the loss of reputation they can suffer from spreading misinformation, even those who agree with their message would not share them widely. In contrast, high homophily creates echo chambers, where users share low-reliability messages aligned with their beliefs, because they understand that there are few negative reputational consequences from doing so. Misinformation contained in low-reliability articles can then spread virally in these echo chambers.

Our framework enables a tractable study of platform incentives in designing algorithms that determine who shares with whom. To do this, we assume that the platform aims to maximize user engagement (which is a good approximation to the objectives of major social media platforms such as Facebook or Twitter). Our main result is a striking one. When an article is highly reliable, the